Audits for Bias in AI Systems

Brenda Leong

Director AI Division
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WHEN AI GOES WRONG

Incidentdatabase.ai

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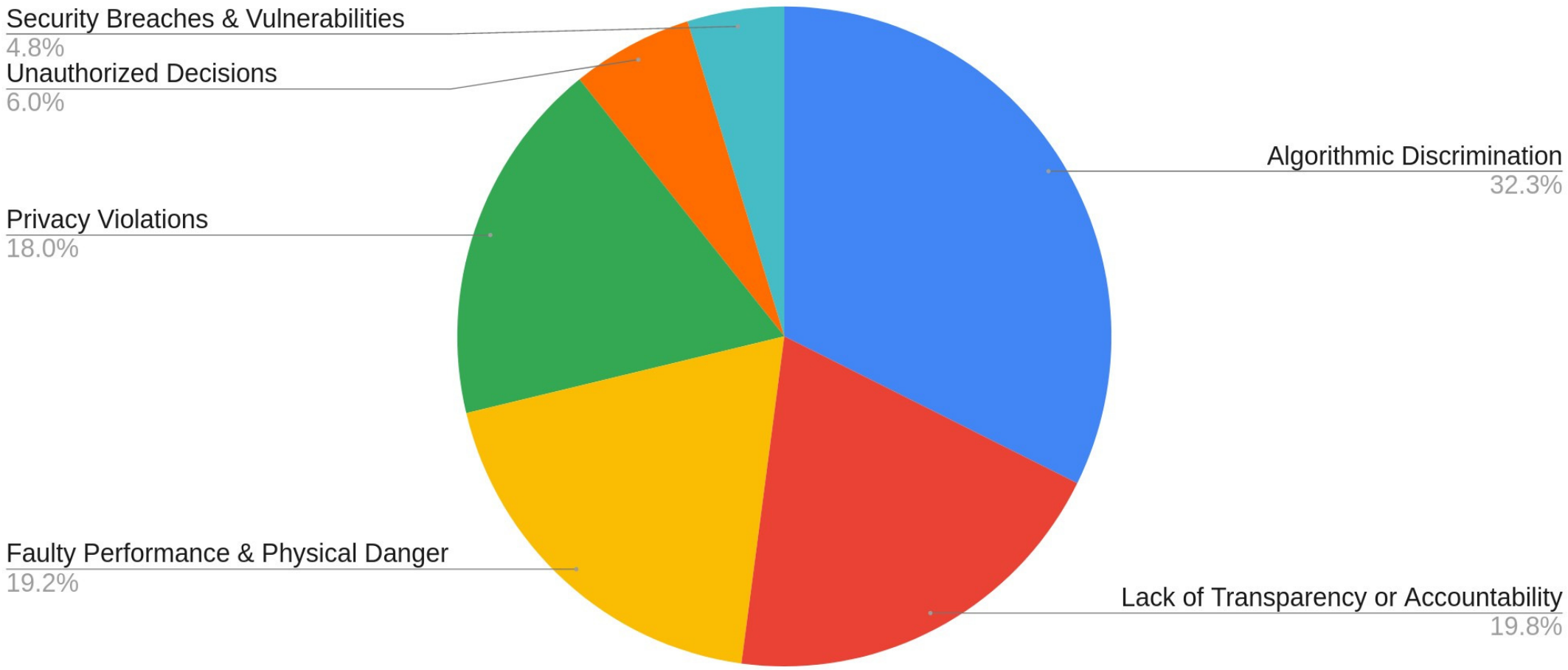
COMMON FAILURE MODES



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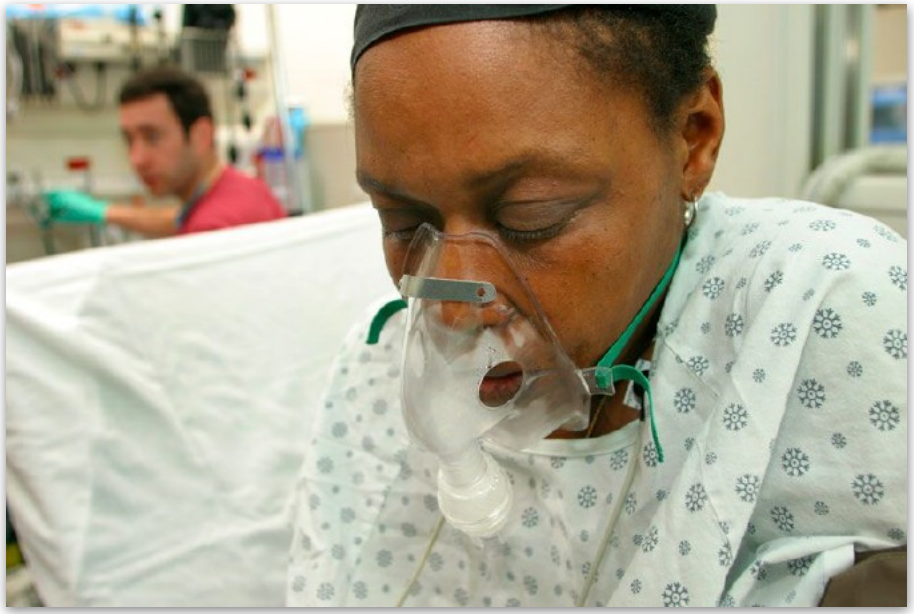
Harms Are Often Concentrated on Marginalized Groups

- Lack of access
  - Screenout
  - Digital Divide

- Bias and opacity in:
  - Credit scoring and credit products
  - Criminal risk assessment instruments (RIAs)
  - “Ed tech”
  - Employment screening systems
  - Fraud detection systems
  - Healthcare resource allocation
  - “Landlord tech”
  - Predictive policing systems



Millions of black people affected by racial bias in health-care algorithms

- Nature, 24 Oct. 2019

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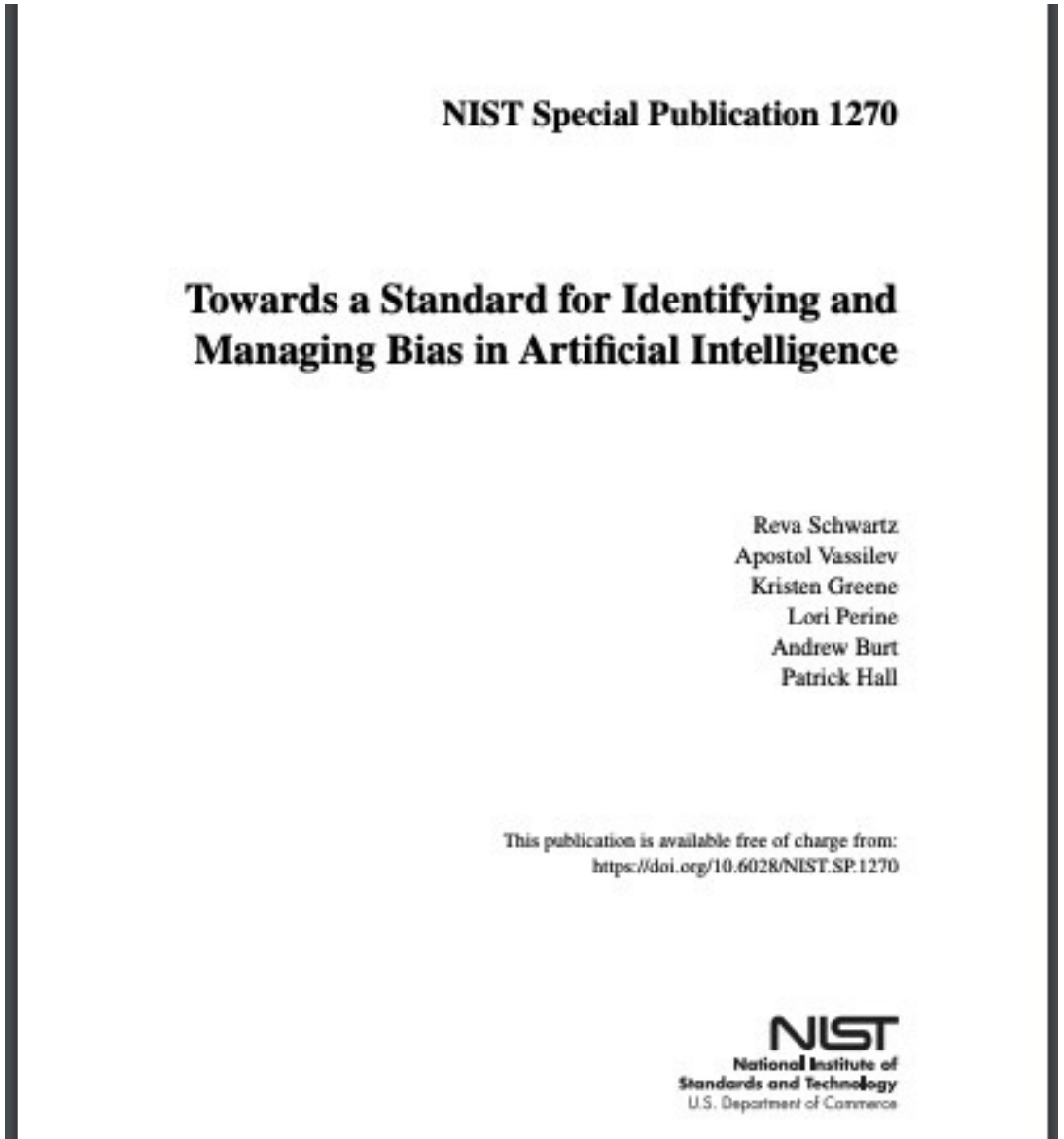
Fairness: Discrimination Testing

Testing for Discrimination:

- Group disparities
  (preferably using tests with legal standing)

- Individual disparities
  - Counterfactuals
  - Track decision boundaries
  - Train adversary models or use special training constraints



NIST Special Publication 1270

Towards a Standard for Identifying and Managing Bias in Artificial Intelligence

Reva Schwartz
Apostol Vassilev
Kristen Greene
Lori Perine
Andrew Burt
Patrick Hall

This publication is available free of charge from:
https://doi.org/10.6028/NIST.SP.1270

NIST
National Institute of Standards and Technology
U.S. Department of Commerce

AI AUDITS

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What is an Audit?

- **Official tracking of adherence to policy, regulation, or law**
  - Legal standards or regulatory frameworks
- **Conducted by independent third parties**
  - Internal, external
  - Transparent and fair
- **Reporting outcomes based on audience and purpose; accountability**
  - Public; regulators; internal overseers
  - Agreed upon standards
  - Certifications
  -

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Measuring Bias

More than data and model outputs

- Unconscious bias and overt prejudice
  - Initial design choices (“screenout”)
  - Homogenous engineers/perspectives
  - Socio-technical applications

Measurement Challenges

- Data labeling
  - Demographic descriptions (BISG)
  - Language limitations
  - Benchmarks
  -

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Tangible Practices

- **Mathematical definitions of bias**
  - Avoid overly complex, sophisticated or unexplainable thresholds
- **Convert outcomes to a binary or single numeric outcome**
  - Apply traditional measures of practical and statistical significance
  - Select appropriate statistical measures (AIR, *t*-tests, Fisher's exact test)
  - Align traditional metrics to existing laws and regulations

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Example: Bias Audit Methodology

- Score each video with FakeFinder models.
- Segment scores by demographic group
  (intersectional groups not considered in this case).
- Establish protected groups:
  East Asian, Black, South Asian, and Women.
- Establish control groups: Whites and Men.
- Test for practical and statistical significance in ***outcome*** differences:
  - **Statistical significance:** *t*-test, significance at $p = 0.05$
  - **Practical significance:** adverse impact ratio (AIR)
- Acceptable threshold: 0.8 – 1.25 (4/5th's rule)
- Ideal threshold: 0.9 – 1.11
- Test for practical significance in ***performance*** differences:
  - **Practical significance:** Accuracy, TP, TN, FP, FN rates
- Acceptable threshold: 0.8 – 1.25 (4/5th's rule)
- Ideal threshold: 0.9 – 1.11

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Example results

- **Practical Significance: AIR**

For every 1000 deepfakes detected with White faces, we expect 694 deepfakes with S. Asian faces to be detected.

Demographic Groups	AIR
E. Asian-to-White	1.004
Black-to-White	0.821
S. Asian-to-White	0.694
Female-to-Male	1.035

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Example results

- **Statistical Significance: t-Tests**

True positive scores for White faces are on average 2.53% lower than for S. Asian faces. This difference is significant, but the actual difference is moderately small. Sample size and a narrow standard deviation for S. Asian scores contribute to the statistical significance, but so does the difference in group means.

Demographic Groups	Control Mean	Comparison Mean	Percent Difference	*p*-value
E. Asian-to-White	0.948	0.964	-1.69	3.39E-04
Black-to-White	0.948	0.926	2.32	6.65E-02
S. Asian-to-White	0.948	0.972	-2.53	4.06E-04
Female-to-Male	0.955	0.948	0.73	1.62E-01

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Example results

- **Performance and Error Rates**

E. Asian faces experience 644% of the false positive rate that White faces experience.

Demographic Groups	Acc. Ratio	TPR Ratio	FPR Ratio	TNR Ratio	FNR Ratio
E. Asian-to-White	1.005	1.012	6.438	0.973	0.394
Black-to-White	0.969	0.951	0.000	1.005	3.488
S. Asian-to-White	1.017	1.020	0.000	1.005	0.000
Female-to-Male	0.988	0.987	#DIV/0!	0.992	2.276

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Example: Audit Conclusions

- **Do deep fake (detectors) discriminate?** Yes, like nearly all other socio-technical AI systems.
- **Bias tests indicate disparity in both outcomes and performance.** Performance ratios point to problems in erroneous decisions. (High-confidence erroneous decisions are a common failure of neural networks.)
- **Biased and wrong deep fake detection consequences.** Bias causes wrong decisions and allows for adversarial exploitation.
- **Remediation via technical or process means is essential.**
- **Analysis via causal or explainable AI (XAI) methods is required to understand drivers of bias.**

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NYC Local Law 144

- **Requirements:** Companies using an automated decision tool in their employment process provide notice, audit the system, and publish results
  - Employers or employment agencies who use such tools must use an external auditor
  - Output tables are provided, in alignment with existing EEOC practices
- **Areas of Contention:** Definitions, required data, relationship between vendor providers and complying employers
- **Milestones:** First formal “AI Audit” law

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RISK & COMPLIANCE JOURNAL

New York’s Landmark AI Bias Law Prompts Uncertainty

Companies that use AI in hiring are trying to determine how to comply with a New York law that mandates they test their systems for potential biases

LOCAL LAWS
OF
THE CITY OF NEW YORK
FOR THE YEAR 2021

No. 144

Introduced by Council Members Cumbo, Ampry-Samuel, Rosenthal, Cornegy, Kallos, Adams, Louis, Chin, Cabrera, Rose, Gibson, Brannan, Rivera, Levine, Ayala, Miller, Levin and Barron. Passed under a Message of Necessity from the Mayor.

A LOCAL LAW

To amend the administrative code of the city of New York, in relation to automated employment decision tools

Be it enacted by the Council as follows:

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COMPLEXITIES

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CONTEXTUAL BIAS

Fairness - what does it mean?

- **Algorithmic Fairness (1) -** Equally accurate for members of legally protected groups (statistical bias)
- **Algorithmic Fairness (2) -** The algo produces the same percentage of false positives or false negatives for each group
- Impossible to maximize both/achieve parity on all axes

Fairness - Conceptual; normative; legal

- **Conceptual -** A score has same meaning across groups - equal predictive accuracy. Not the best measure?
- **Normative -** Parity in ratio of FP/FN is normatively significant, indicative of likely unfairness
- **Legal:** The collection and use of data on protected classes is necessary for testing across either standard.

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TECHNOLOGY IN CONTEXT

- **Bias in Social Process**
  - Randomization
  - Predefined constraints
- **Design and Practice**
  - Watch for discriminatory outcomes in actual practice
  - Connect to context and policy controls (design of emergency services; policing routes;

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REGULATORY APPROACHES

- **Risk Assessment**
  - Future oriented
  - Take risks, and avoid harms
- **Policy Baggage**
  - Threshold question - Normative choice to use technology
  - Autonomy, dignity, privacy, equality, and human rights
- **EU AI Act**
  - Unacceptable risk → Prohibited Use
  - Values based, not technology effectiveness

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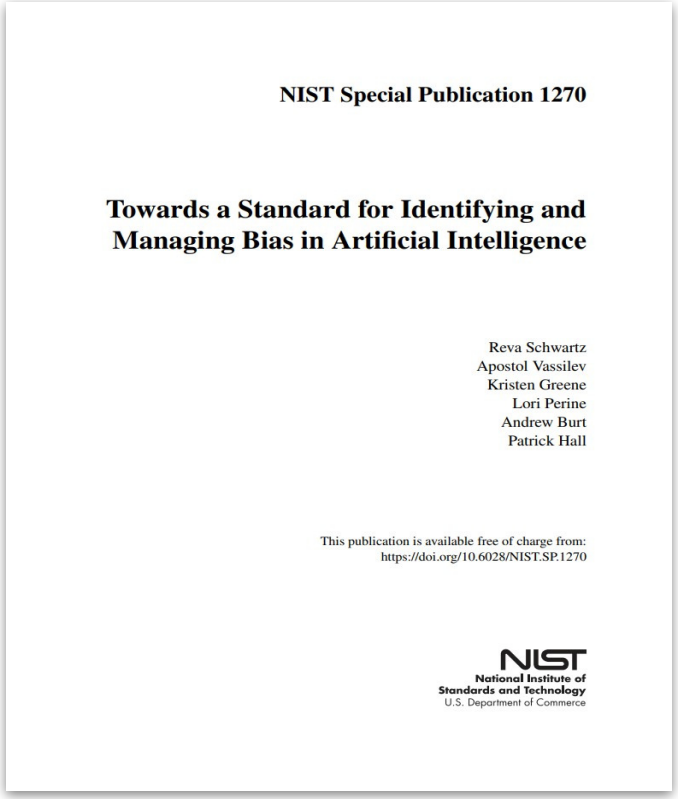
NIST - Governance Practices

NIST SP 1270 Bias Guidance



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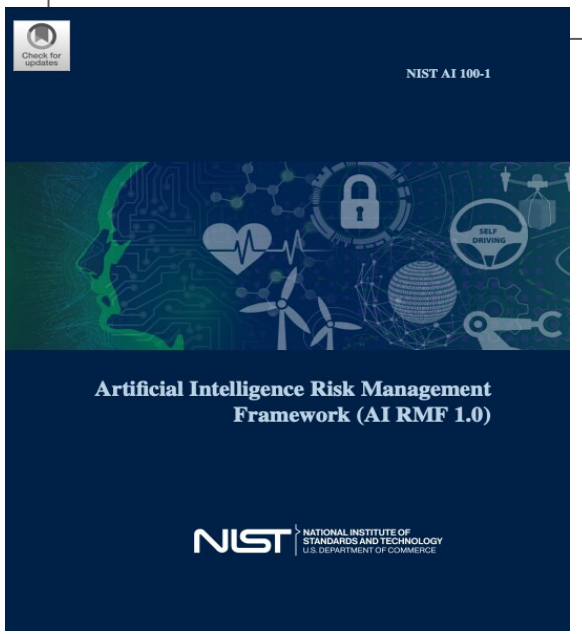
National Institute of Standards and Technology
U.S. Department of Commerce

NIST AI Risk Management Framework

Core RMF: high-level perspectives and guidance

RMF Playbook: lower-level guidance and resources.

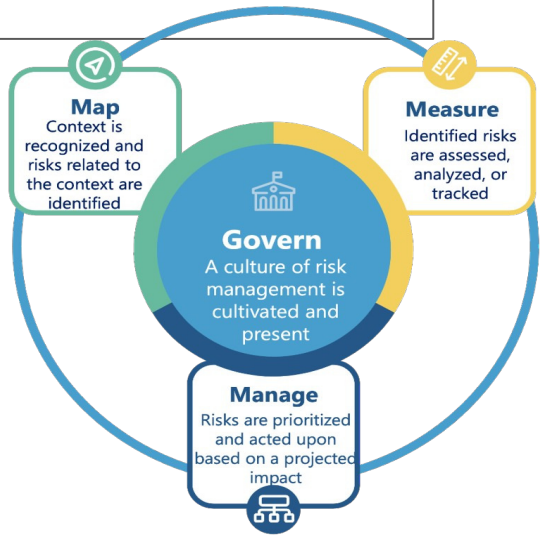


Check for updates

NIST AI 100-1

Artificial Intelligence Risk Management Framework (AI RMF 1.0)

NIST NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY U.S. DEPARTMENT OF COMMERCE



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Model Audits - Summary

- **Audits may have different meanings in the context of AI**
- **To what standard do we audit?**
  - Legal and industry standards
  - Who audits the auditors?
- **Audits are being incorporated** into regulatory and legal oversight systems

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THANK YOU -

QUESTIONS?

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Ethics of AI
Prof. Evan Selinger
Unit #7 AI Ethics and Audits

Due Date
All of the assignments for this unit are due by the end of the day on Sunday, December 7th.

Main Goal
The main goal of this unit is to understand the emerging field of AI audits—what they are, why they matter, and the complex challenges involved in implementing them effectively. By examining real-world examples and expert perspectives, students will gain insight into how auditing serves as a critical tool for responsible AI governance, while also recognizing its limitations and the ongoing debates about best practices.

AI Audits: A Complex and Evolving Field
The previous unit on AI ethics in hiring and work demonstrated that AI systems can cause significant harm when they exhibit bias, operate inaccurately, or fail to respect human dignity. A natural response to these problems is to audit AI systems—but what exactly does that mean?

As Brenda Leong, Director of the AI Division at ZwillGen PLLC, reveals in her extensive work with corporate clients, AI audits are far more complex than they might initially appear. Unlike financial audits, which have decades of established standards and practices, AI audits are a relatively new field with significant definitional challenges and practical complications.

Learning Objectives
Understand the challenges in defining what constitutes an AI audit, recognize the key implementation challenges, including data selection, auditor independence, and frequency decisions, evaluate the strengths and limitations of New York City's Local Law 144, assess the audit washing concern and arguments for why audits remain valuable despite this risk, analyze the role of values and judgment in technical measurements, and understand how AI audits fit into broader frameworks for responsible AI governance.

Watching and Reading Assignments
1. Watch my interview with Ms. Leong where we discuss the foundations of AI auditing and key implementation challenges. You can also download a written transcript of it.

2. Read Brenda Leong, Evan Selinger, and Albert Fox Cahn's "AI Audits: Who, When, How... Or Even If?" Mr. Fox Cahn is the founder and executive director of the Surveillance Technology Oversight Project.

3. Watch the lecture on AI audits that Ms. Leong delivered to my in-person Ethics of AI course.

Writing Assignment
4. Write Reflection Paper #5. Reflection Paper assignments are located in the Assignments section of MyCourses.

AI Audits:
Who, When, How…Or Even If?

Evan Selinger, Brenda Leong, and Albert Fox Cahn

"One must confront the reality that audit discourse tends to be inexact and confusing. There is no settled understanding of what an 'algorithmic audit' is." – Ellen P. Goodman and Julia Trehu. (2022, 3)

Introduction

Artificial intelligence (AI) tools are increasingly being integrated into decision-making processes in high-risk settings, including employment, credit, health care, housing, and law enforcement. Given the harms that poorly designed systems can lead to, including matters of life and death, there is a growing sense that crafting policies for using AI responsibly must necessarily include, at a minimum, assurances about the technical accuracy and reliability of the model design.

The parameters of appropriate design are greatly debated. (Narayanan 2018).[1] A frequently expressed position is that AI should not be designed with features that reinforce and amplify the adverse social and economic disparities that have most often impacted historically marginalized and minority populations. (Benjamin 2019). Determining if any particular system perpetuates these harmful inequities is not simple, but public and private sector actors are developing rules and standards for conducting AI audits that test whether an AI programming perpetuates these problems and other matters of injustice. (Schaake and Clark 2022; Wilson 2020).[2] The effort to streamline and advance AI audits includes contributions from government agencies, international standards-setting bodies, academics from a range of disciplines, advocacy groups, and a host of commercial companies.

Reaching an agreement on those rules and standards has thus far proven quite challenging. Far from having a single definition, even the concept of an "audit" has different meanings in the AI literature. Definitions range from broad notions that cover "any assessment of an algorithm" to more restricted interpretations that parallel the financial concept of an audit. (Hasan and others 2022, 4). When we begin our discussion of AI audits, we will use the term in the latter sense; doing so matches how many, but certainly not all, AI audit practitioners speak. From this perspective, "an audit is an independent assessment or evaluation of an organization's algorithm, using transparent rules or laws" for the purpose of serving "society, the public, users, or some other body independent of the user's organization." (Hasan and others 2022, 4).

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[1] For starters, if appropriate design includes considerations of justice, then one must acknowledge, from the start, that there are a range of definitions of "fairness," and the definitional differences can have significant political consequences.

[2] Stanford University's 2022 AI Audit Challenge provides a useful summary of the benefits of AI auditing, namely verification, performance, and governance.

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On a more granular level, this definition of audit specifies the review process which produces a particular type of output: a "binary" one. (Hasan and others 2022, 4; Bhaumik and Dey 2022). Unlike modes of assessment that provide qualitative evaluations or assurances, this type of AI audit tests whether a system operates as intended with regard to fairness or equity for identified categories or demographics. These analytics can then be compared to existing standards to determine if the system complies or fails to comply with relevant "rules or laws." (Hasan and others 2022, 4). From this perspective, an AI audit is primarily a technical operation rather than a situational one, even though every system in the real world operates in a social-technical context where people make decisions about how to design, use, and test technology. While the technical process of an AI audit must at least produce a testing output, it also can contribute related services, such as proposing technical mitigation measures for debiasing models and reducing risk. (Jones and others 2023).

Just as "audit" has many meanings, so does "AI." In this chapter, we focus on a specific variety of machine learning systems. When we refer to AI, we mean machine learning models that analyze patterns and perform a predictive function to produce quantitative outputs, scores, or rankings. We have restricted our purview to this type of machine learning because it is the most common model currently used in commercial applications and is at the center of most policy discussions of AI audits. To be sure, one can audit other types of AI, such as *reinforcement learning* systems that do not require training or testing data, and *generative AI* that takes in enormous amounts of data and creates new content in the form of images and audio (e.g., deep fakes, Midjourney, and WOMBO) and text (e.g., ChatGPT). However, compared with predictive machine learning algorithms, there is even less developed operational guidance for responsibly assessing these systems. Furthermore, generative AI technology currently only has limited real-world applications, despite the flurry of media coverage.

Because AI auditing is still in its early stages, many questions remain about how to best conduct them. While many people are optimistic that valid and effective best practice standards and procedures will emerge, some civil rights advocates are skeptical of both the concept and the practical use of AI audits. These critics are reasonably concerned about audit-washing—bad actors gaming loopholes and ambiguities in audit requirements to demonstrate compliance without actually providing meaningful reviews. (Goodman and Trehu 2022, 3).

They also reject the narrow definition of an audit found in much of this chapter (as one part of a larger oversight process). Indeed, they point out that technical audits can backfire by giving an institutional seal of approval to tools that can still be used to harm vulnerable communities disproportionately. In other words, critics believe the potential harms resulting from some AI tools are too dangerous to risk allowing into society regardless of what audits of system performance reveal; they warn that the very existence of audits may help politicians and regulators normalize oppression, given the perceived objectivity of the testing.

This chapter aims to explain why AI audits often are regarded as essential tools within an overall responsible governance system and how they are evolving toward accepted standards and best practices. We will focus most of our analysis on these explanations, including recommendations for conducting high-quality AI audits. Nevertheless, we will also articulate the core ideas of the skeptical civil rights position. This intellectually and politically sound view

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should be taken seriously by the AI community. To be well-informed about AI audits is to comprehend their positive prospects and be prepared to address their most serious challenges.

One of the primary reasons why AI audits are contentious and difficult to conduct is that they are a new endeavor and lack the type of professionalization standards that emerge when a community of practitioners matures. By contrast, financial audits–which are an essential regulatory mechanism–have evolved over nearly a century. We believe the financial counterparts are a useful baseline comparison for thinking about the future of AI audits, and we begin by articulating why–emphasizing the auditor's obligation to perform their job with due care against objective standards. Then we explain why New York City recently passed a first of its kind AI audit law and identify its fundamental promises and limitations. Our analysis presents enough detail about the law to treat it as a case study. In general, case studies are a valuable tool for figuring out how to address complex problems because they offer nuanced detail of sticking points. In this instance, the case study approach helps clarify why the maturity level of today's AI audits might be likened to the state of 1950s financial audits. The process is far from arbitrary and subjective, but it still has a long way to go.

How, then, can AI audits be designed to meet the task at hand, which is to support overall responsible AI governance standards and practices? We answer this question in stages that collectively provide guidance. First, we propose three foundational claims that limit the scope of what counts as an AI audit. Second, we group the most critical issues for conducting reasonable AI audits into five topics, each framed with a guiding question or questions. Third, we revisit the New York City audit requirements and evaluate them through the lens of our guiding questions. We clarify what that audit emphasizes and, based on some of its limitations, identify ways that future audit guidance can be improved. These recommendations complete our pro-audit evaluation and guidance. Then, we present the anti-audit position. It articulates civil rights-based skepticism of many of the core claims made in the chapter. Finally, we offer concluding remarks and present key learnings.

A Brief History of Financial Audits as a Regulatory Mechanism

Auditing financial systems is a well-established and recognized regulatory process for examining the accuracy and adequacy of a company's financial information. These audits are performed in accordance with generally accepted accounting principles and within the applicable regulatory requirements in the relevant jurisdiction. (U.S. Department of Justice, Office of Justice Programs 2022). The trained, professional auditor aggregates and evaluates all the information provided or derived to determine whether the financial statements adequately represent the position of the company. In short, auditors review selected information and formulate a judgment. While quantitatively based, financial auditing still somewhat reflects a qualitative evaluation of the relevant data and the financial reporting of the organization. Auditors, whether external or from independent, internal teams exercise professional judgment and rely on training, experience, industry knowledge, and their own expertise and understanding. (International Organization for Standardization 2016; U.S. Securities and Exchange Commission 1998; Joyce 1976, 29-60).

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Current financial auditing practices have been evolving for over 100 years and are continually adapting to modern circumstances. (Boersma and others 2022). Prior to the 1930 enactment of federal securities laws, the specifics of an audit were entirely at the discretion of the auditor and most audits focused on fraud detection. (Levy 2022). In those days, there was no authoritative body that determined comprehensive accounting standards. Even the term "generally accepted accounting principles" lacked a tangible reference until 1959 when the Accounting Principles Board set out to create coordinated principles to serve as a framework. In short, the validity, accuracy, and standardization of financial audits have adapted and developed over time.

Financial audits are not perfect, but without independent and competent auditors, many fraud cases worldwide would never have been detected. A high standard of professional ethics, as well as levels of industry oversight and regulatory controls are in place to monitor the performance of financial auditors. Although, of course, questions remain as to how well these standards are observed in practice. One code of professional conduct states that auditors must go about their business with "due care." In law, due care is related to the concept of how a "prudent person" behaves, and it has four implications for what auditors should know and how they should conduct themselves. (Schmidt 2023). These professional expectations could certainly be applied to developing guidance around "who" should audit AI systems, as we discuss below.

Auditors Should:	…possess the necessary *skill* to evaluate financial statements
	…follow the *duty* to use reasonable care and diligence
	…work in *good faith* and with integrity while not assuming to be infallible
	…accept *liability* for negligence, bad faith, or dishonesty but not for mere errors in judgment

Figure 1: Auditor Professional Responsibilities (Schmidt 2023)

The evolution of financial auditing continues in the 21st century. The auditing scandal around Enron in 2001 led the Financial Accounting Standards Board to issue a new standard and also led to a major piece of federal legislation that substantively transfigured the auditing world: the Sarbanes-Oxley Act of 2002 (Webber 2022). Among other things, this act introduced new rules around auditor independence, established the Public Company Accounting Oversight Board, and required certain parties to audit and report on the effectiveness of their internal controls over financial reporting.

While AI audits involve a number of considerations that do not apply to financial audits, the brief history provided here suggests the auditing process can evolve with sufficient planning and commitment as standards develop and practitioners professionalize.[3]

[3] For comparison, under the draft European Union AI Act expected to pass by next year, EU member states will create national authorities to collect and analyze information related to AI deployment and conduct periodic audits. Companies will be asked to register with the member-state authorities. The registration requires a risk assessment

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New York City's Pioneering AI Audit Law

Our detailed discussion of AI audits begins by examining a novel New York City law that addresses automated decision-making systems when used in employment practices (New York City Council 2021B). To grasp the impetus for the law, it is necessary to understand the underlying context. Companies are progressively integrating these automated systems into their hiring and promotion processes. (Schellmann 2021). According to a February 2022 survey from the Society for Human Resource Management (2022), nearly one in four companies rely on automation or AI to "support human-resource-related activities." As the economy recovers from the impacts of COVID-19 and the resulting employee turnover partly associated with people reassessing their professional and personal priorities (also known as the "Great Resignation"), many companies see evolving AI-based technologies as important infrastructure for increasing efficiencies and lowering costs.

Unfortunately, many employment professionals have limited experience with these new systems and do not understand the main risks associated with using automation for human resource (HR) functions, such as resume screening or applicant rankings for raises and promotions. Given the technological novelty, lack of associated training, and varying company practices and expectations, many HR managers are not well-versed in technology-centric concerns such as how various configurations of training and testing data can lead to biased outcomes or why a lack of transparency surrounding the use of AI can create due process issues.

In 2021, New York City addressed this situation by becoming the first U.S. jurisdiction to pass a law regulating these issues within the city's boundaries. It is an instructive case study for two reasons. First, as novel legislation, lawmakers could not structure their rules around any existing precedents for government-mandated AI audits. Consequently, the specifics of the law have engendered widespread discussion that reflects both areas of agreement as well as clear points of contention that can influence future AI audit laws, even for models that apply beyond the context of employment. Second, while Local Law 144 has an apparently narrow purview, its execution has turned out to be quite complicated. The debate over its implementation highlights the challenge of structuring even a relatively simple AI audit.

Overall, Local Law 144 aims to determine if employers using automated employment decisions tools (AEDT) are generating biased outputs that may affect hiring at the applicable companies (more on this below). To meet this goal, the law imposes four fundamental requirements. First, the law requires employers to have an AEDT audited for bias within a year prior to using it for employment-related purposes, and to repeat such audits at least annually. The law defines an AEDT as "any computational process, derived from machine learning, statistical modeling, data analytics, or artificial intelligence that issues simplified output, including a score, classification, or recommendation, that is used to substantially assist or replace discretionary

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and statement of conformity with other applicable law. Companies utilizing AI tools will be under an ongoing obligation to provide information to the national authorities, including notifications of breaches or incidents, as well as new risk assessments performed and reported in response to major changes to the tool. High-risk AI systems, as defined in the law, will be registered and published in an EU-wide database. Operators of these systems are also subject to heightened scrutiny of their source code, which must be provided to regulators on request.

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decision making for making employment decisions that impact natural persons." The law specifies that an AEDT does not include tools such as a junk email filter, firewall, antivirus software, calculator, spreadsheet, database, data set, or other compilation of data.

Secondly, the law specifies that the audit must be an "impartial evaluation by an independent auditor" and includes testing for two of the existing Equal Employment Opportunity Commission protected categories: race/ethnicity and sex. (42 U.S.C. § 2000e-8(c); 29 CFR § 1602.7; U.S. Equal Employment Opportunity Commission 2022).[4] The reviews of those protected classes already required under existing law are to protect against the legal measure of discrimination called disparate impact. Disparate impact occurs when policies, practices, rules, or other systems that appear to be neutral result in a disproportionate effect on a protected group. For example, having a height requirement for job applicants might appear to be neutral. However, it leads to fewer women being hired as they are less likely, on average, to meet it. In an automated system, disparate impact might be based on a standard screening test for all employees that routinely results in more whites scoring above a threshold score than non-whites. The law aims to protect New York City employees from this type of discrimination by structuring audits of AEDTs around an existing regulatory standard for measuring disparate impact called an *impact ratio*. Federal courts and agencies typically use this standard when dealing with disparate impact in employment contexts, but states and localities have yet to adopt this standard in assessing compliance with state and local employment law. (N.Y.C. Department of Consumer and Worker Protection 2023.[5]

Third, the law requires advance notice to applicants that AEDT systems are in use during the employment process and specifies what the notice must include, including details about what information the automated function collects or uses. Finally, the audit report—the table of demographic results required by the law—must be made publicly available online. The information must also include the date of the most recent audit and the source and explanation of the data used."[6] Since the law is designed to provide transparency as to the performance of the systems being used, the legislation does not address more radical goals. Some hope that regulating AI in the employment sector will prove to be a tool to comprehensively address historical injustices. However, the NYC law does not have that sort of systemic overhaul as part of its agenda. (Coeckelbergh 2022, 42-48).[7]

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[4] The Equal Employment Opportunity Commission is the federal agency charged, among other things, with enforcing anti-discrimination laws in the workplace. These established categories are already required to be reported by employers pursuant to § 2000e-8(c) of Title 42 of the United States Code as specified in Part 1602.7 of Title 29 of the Code of Federal Regulations.

[5] In its guidance, NYC has proposed that '"Impact ratio" means either (1) the selection rate for a category divided by the selection rate of the most selected category or (2) the scoring rate for a category divided by the scoring rate for the highest scoring category'. The guidance can be found in the references.

[6] Specifically, Local Law 144 requires employers to provide notice of use of an AEDT to candidates and employees at least 10 days prior to its use." The notice should also include the job qualifications and characteristics the AEDT will use in its assessment and, if not already made publicly available, "information about the type of data collected for the [AEDT], the source of such data and the employer or employment agency's data retention policy ….". The guidance expands on these requirements.

[7] For more on the differences between compliance oriented and prioritization approaches to employment discrimination, see *The Political Philosophy of AI* by Mark Coeckelbergh in the references.

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Intersectional Categories					
			# of Applicants	Scoring Rate	Impact Ratio
Hispanic or Latino		Male	16	75%	1.00
		Female	12	50%	0.67
Non/Hispanic or Latino	Male	White	20	35%	0.47
		Black or African American	20	50%	0.67
		Native Hawaiian or Pacific Islander	4	75%	1.00
		Asian	12	58.3%	0.78
		Native American or Alaska Native	8	62.5	0.83
		Two or More Races	12	50%	0.67
	Female	White	20	40%	0.53
		Black or African American	12	50%	.67
		Native Hawaiian or Pacific Islander	4	50%	0.67
		Asian	12	25%	0.33
		Native American or Alaska Native	8	62.5%	0.83
		Two or More Races	8	50%	0.67

Figure 2: Sample table as provided in NYC Proposed Rule[8]

New York City lawmakers targeted employment for several reasons. As discussed above, the use of these automated systems by employers is expanding quickly. Employment decisions can have a significant impact on our personal lives while also revealing how well or poorly society addresses social prejudices. Even though it is well known that humans can be biased and that traditional hiring and promotion processes can be unfair, people are still concerned about being judged by an artificial program.

These reservations are well-founded. Algorithms that determine who is shown targeted ads for specific jobs have displayed age and gender-related biases. (Lobel 2022, 49-52). Automated resume screening software has demonstrated gender bias, and people are worrying about how to write resumes that can make it past algorithmic gatekeepers. (Lobel 2022, 57-58). Companies are testing candidates' cognitive abilities and emotional attributes with video games that have the potential to be biased against disabled workers. (Brown and others 2020). Deep uncertainty exists about whether companies are using AI to make layoff decisions and, if so, whether their machine learning systems are calculating and including uncertain future possibilities–such as which individuals are most likely to quit. (Verma 2023). Even more worrisome, job candidates have been examined by software using unproven and opaque tools that "analyze voice, tone, and behavior along with word choice and facial movements" to "screen for qualities such as grit, curiosity, and polish," (Lobel 2022, 61) even though such systems have

[8] This table is accompanied by the following explanation: "(b) Where an AEDT scores applicants or candidates, a bias audit required by § 20-871 of the Code must, at a minimum: (1) Calculate the average score for individuals in each category; (2) Calculate the impact ratio for each category; Example: An employer uses an AEDT to score applicants for "culture fit". To do so, the employer must ensure that a bias audit of the AEDT was conducted within a year from the planned use of the AEDT. The employer uses historical data on average "culture fit" score of applicants for each category to conduct a bias audit as [in the table]:"

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yet to be proven scientifically, and indeed have been shown to lack validity. (Barrett and others 2019, 1-68).

Despite the perceived need to enact laws controlling AI in such fraught contexts as employment, such as Local Law 144, responses to its passing have been mixed. (Maurer 2021).[9] The most prominently debated issues during the public review process included the specific choices made for what to test for, what standard to apply, what data to use, what to include in the notice and the public post of the audit report, how to define terms, and even what "substantially assist" means, as it impacts what systems might be covered by the law and therefore require an audit at all. Additionally, much debate centered on the law's limited scope for addressing bias. On the one hand, the proposed guidance for implementing the NYC law goes further than existing federal employment protections because it requires intersectional analysis (e.g., analysis that compares black women/black men as well as blacks to whites overall). On the other hand, many complaints expressed in the public comments and feedback to the draft guidance emphasized the importance of conducting audits that cover all protected classes (i.e., expand to include religion, sexual orientation, gender identity, pregnancy, national origin, age, and disability). (Employment Opportunity Commission 1979: 1179-12015).[10]

In other words, there is a spectrum of concerns around the proposed testing via these audits. One way to view this current limitation to specific classes is through the lens of pragmatism: given the constraints, the perfect should not be the enemy of the good; the audit needs to begin somewhere, and lawmakers can subsequently add additional testing parameters. However, another perspective might conclude that abridged justice is not justice at all because partial protection means illegitimately assigning more worth to some protected classes over others; this is unacceptable because each one has an equal claim to fair treatment. From this perspective, the standard for passing this mandated AI audit falls below the standard that justice requires. But also, there is the question posed by some, as addressed below, as to whether audits can ever be a useful tool for responsible oversight.

These sticking points are not surprising, but it will be important to reach some level of consensus around solutions, even if those may allow for variations based on the risk and context of different applications. Several states and municipalities are developing laws that require AI audits and will look to NYC as a touchstone to structure further deliberation. (N.Y. State Senate Bill S6852 (2022); N.J. Senate Bill 1926 (2022); C.A. Civil Rights Council Proposed

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[9] Julia Stoyanovich, a professor of computer and data science at New York University and the founding director of the school's Center for Responsible AI, sees the law as a "substantial positive development," especially the disclosure components informing candidates about what is being done, contrasted with The Center for Democracy and Technology in Washington, D.C., which called it a "deeply flawed" and "weakened" standard that doesn't go far enough to curb AI technology bias in employment.

[10] While a broader scope is ideal, the reality is that the law's limited jurisdiction (i.e., scoped only to affect jobs within New York City) creates statistical constraints. While more data can be collected, the numbers are unlikely to be sufficiently representative. At the national reporting level, the Equal Employment Opportunity Commission has created an exception for this circumstance so that small samples aren't included in the mandatory reporting because they are not statistically meaningful, and the NYC law also reflects such an exception which may preclude small populations or smaller groups of employees entirely. It is also important to note that other commenters were opposed to the idea of audits altogether and felt general restrictions on use, or other controls and limitations would be more appropriate at the current state of technology.

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Modifications to Employment Regulations Regarding Automated-Decision Systems (2022); D.C. Bill 24-558 (2021); Commonwealth of Massachusetts State Senate Bill S.D. 1827 (2023)).[11] Since Local Law 144 passed, companies, advocacy groups, and other stakeholders have requested further details specifying how to meet compliance conditions. The NYC Department of Consumer and Worker Protection proffered two drafts of specific guidance covering key areas and held two public hearings to allow for comments, culminating by issuing final compliance guidance on April 5, 2023 (New York City Department of Consumer and Worker Protection 2023). In addition to those already discussed, the NYC rules include the following requirements.

- *Data requirements*: The proposed implementation guidance for this law stipulates that these bias audits must use the historical data (i.e., the data generated by the employer's real-world use of the system) of the AEDT being audited unless there is not enough such historical data available. In that case, either aggregated or test data may be used.[12] Test data—the data used in the original design and development of the system—is more easily obtained and measured. However, it may not reflect how the model has actually performed in the real world.

- *Independent auditor*: The current guidance defines an independent auditor as anyone who is not an employee of the vendor or the employer, was not involved in using, developing, or distributing the AEDT, and does not have a direct or materially indirect financial interest in the employer or the vendor.

Despite the clarifications provided by the evolving guidance on many of the points of concern, there remains significant uncertainty among employers trying to prepare for compliance. One remaining ambiguity is the failure to address remediation for subpar results. That is, while this law simply requires disclosure of audit performance, the output is not to be measured against any existing standard. The employer would be held to disparate impact standards under existing law, but the NYC law does not address what will happen if the audit outputs do not meet established disparate impact requirements. Under federal equal opportunity employment practices, vendors or employers would typically be allowed to take remedial action to correct such outcomes or to provide a business justification for the out-of-bounds outcomes when the ratio results fall below established standards. The NYC law does not address this necessary follow-up of a bias evaluation.

Where do the unresolved and debated issues leave us? Local Law 144 is an instructive case study because its sticking points–only some of which were highlighted–show that audits are tools that have to be designed for their particular context and purpose, and for which there will almost never be general agreement when it comes to the details. Like earlier financial audits that occurred before generally accepted accounting principles were formed, the AI audit process has a long way to go.

The Challenges and Realities of Auditing AI Systems

[11] It is worth noting that some, though not all, of these pieces of legislation are focused on the employment context.

[12] Test data is not well defined. The rules also provide that where test data is used, the audit report must include an explanation as to "why historical data was not used and describe how the test data used was generated and obtained."

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Considering the history of financial audits, the New York City case study, and expert insights on responsible algorithmic governance, we can now move the analysis forward and begin to answer a critical question: *What does a reasonable AI audit look like?* The next methodological step is to establish foundational claims.[13]

As noted in the Introduction, there are many definitions of "audits." To determine a structure for reasonable AI audits, we recommend, as our *first foundational claim,* adopting the narrowly circumscribed one that aligns with financial audits. Since financial audits are "an official examination of an organization's accounts by an independent body," an AI audit is the official examination of a model's performance metrics. That is, an AI audit specifies whether a model's performance metrics comply with a rule or law or fail to align with it. It also can include related technical analysis, such as identifying technical solutions for de-biasing models and reducing risk.

Many companies or regulators are likely to adopt this definition of an audit, and it appears to closely align with the intended meaning of audit in NYC's Local Law 144 and other proposed legislation. While the National Institute of Standards and Technology's (NIST) recently published the AI Risk Management Framework for overall AI governance speaks more about broader "assessments," when it does refer to "audits," it does so in the same narrow capacity that we have selected. (National Institute for Standards and Technology 2023). By any measure, NIST implies that audits are a step within or aspect of a more extensive governance process.

Acknowledging this part/whole relationship leads us to formulate our *second foundational point*. That is, within the multi-faceted context of responsible AI governance, an audit is a necessary but insufficient part of a management or regulatory framework. One of the limitations of NYC's Local Law 144 is that it both begins and ends with the audit, creating new requirements for employers using AEDTs but without imposing additional accountability for how AEDT systems are selected or applied, most obviously by failing to address the responsibility of vendors who may sell biased tools. Determining the boundaries of risk assessment, accountability, and contextual oversight are important governance issues even though they fall outside the scope of what the audit itself includes. Consider: Auditing a system for bias by statistically testing and analyzing its outputs is critical for determining if the system competently performs the function it is programmed to carry out. (Zhang and others 2018). However, that evaluation does not address the overall risk of the system.

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[13]In the European Union, the Digital Services Act is important legislation impacting audits, but applies only to "very large online platforms" (VLOPs). The DSA is based on a model of "co-regulation," and requires an annual internal risk assessment and external independent audit, but on voluntary standards. They do require mitigation measures, although as yet undefined. The expectation is that industry will develop a code of conduct or best practices for addressing certain systemic risks. There is also a requirement that these platforms publish an annual comprehensive report of the most severe systemic and recurring risks of their platforms. While the legislation has specific guidelines on who the auditors should be, and a very general minimal framework for the audit reports themselves, the DSA does not prescribe the audits. It explicitly says that it supports the building of voluntary—presumably industry-based—standards for said audits. Relatedly, the EU's General Data Protection Regulation (GDPR) requires that organizations be able to explain their algorithmic decisions.

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The full array of risks and benefits of an AI system occur in the real world and are based on "the interplay of technical aspects combined with societal factors related to how a system is used, its interactions with other AI systems, who operates it, and the social context in which it is deployed." (National Institute for Standards and Technology 2023). Consequently, the technical evaluation of a system's accuracy and fairness through an audit can never be sufficient to validate the larger question of whether the system is being responsibly managed overall or if organizations, companies, or individuals should even be using it. The NIST framework recognizes the limited role of technical verification by identifying four central components of responsible AI: map (i.e., establish the context to frame risks related to an AI system), measure (i.e., use quantitative and qualitative tools and techniques to analyze, assess, benchmark and monitor AI risk and related impacts), manage (i.e., treat identified risks and decrease the likelihood of system failures and negative impacts.), and govern (i.e., cultivate and implement policies and procedures that create a comprehensive risk management culture). (National Institute for Standards and Technology 2023). Audits only address aspects of two of these processes: map and measure.

In other words, while audits can use data science and statistical analysis to verify compliance with fundamental standards of legal fairness for automated decisions, they are not comprehensive, holistic, and interdisciplinary assessments that identify all of the relevant risks and harms. (Solove 2023). They can still fail to detect essential attributes of a system and decisions surrounding its use that might or will contribute to adverse real-world impacts. For example, a facial recognition system could be audited for bias, and the audit would focus on a performance measure such accuracy rates across demographic groups. However, even if the technology passes the audit standard for fairness, one could still have ethical and political reasons for believing that the technology is too dangerous for society and should be banned.

Relatedly, governance activities that go beyond auditing processes determine which AI systems get audited and when and how. In other words, an audit can be part of the response to a perceived need when authoritative decision-makers, such as lawmakers and business executives, take action. In the case of NYC's Law 144, the perceived need was determining whether AEDTs are introducing unfair bias into the employment decisions of those companies using them. That perceived need could have been ignored; political forces could have led to a different outcome, like waiting for another jurisdiction to take action first, or by banning the use of the systems altogether.

This brings us to our *third foundational claim,* which is about intent. The choice of imposing an audit to address a perceived need must serve a clear and distinctive purpose. The main goal of an AI audit, at least by legislators, should be to ensure that public interests–of individuals, groups, and society as a whole–are protected by making systems and their operators more accountable. In addition to NYC Law 144, other proposed legislation conveys this point (European Commission 2022). This stipulation means that audits are a normative enterprise, and therefore the standards they should be held to, especially when no existing legal or industry standard applies, must be identified or created to address potential harms. Since comprehensive identification of harms requires stakeholder participation, the debate over the parameters for an AI audit should include the various public groups whose interests and rights are supposed to be protected.

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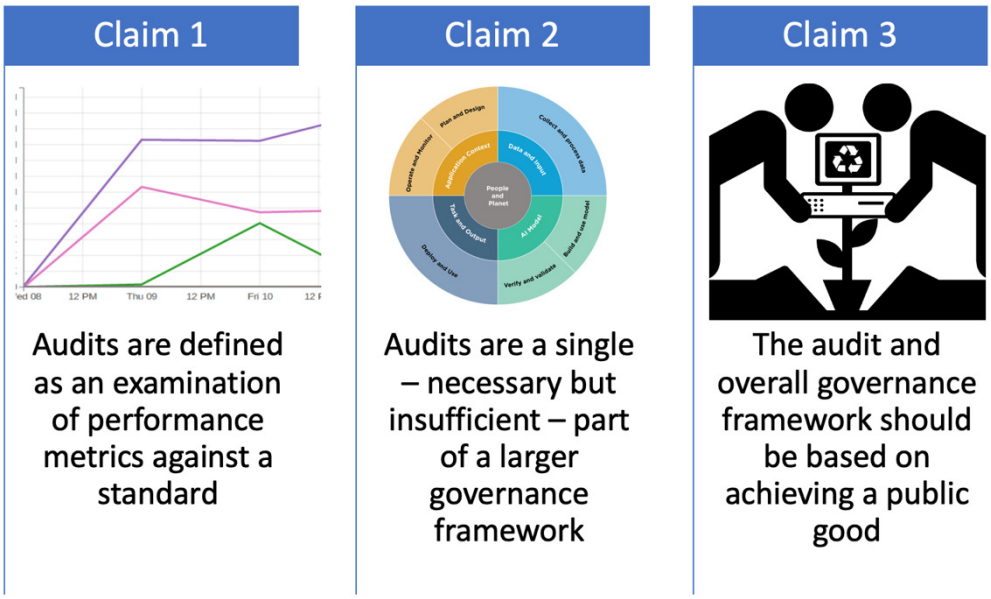
Therefore, an audit is: 1) an examination of a model's performance metrics, 2) as part of a larger governance framework, 3) that is designed with a specific intent to support a public good.



Our three foundational claims thus create a limited scope for AI audits and establish the boundaries driving preliminary guidance. Many thorny issues, however, remain. As the relatively basic expectations of the NYC Law 144 demonstrated points of debate and challenging questions remain. To flesh out this next level of detail defining a reasonable AI audit, we have grouped the most critical issues into five topics that are introduced by a guiding question or questions. For some, we offer suggestions for possible answers, but in other cases, we identify the sticking points that simple prescriptions cannot resolve.

First, *who should perform the audit?* This question will likely have a different answer in different contexts (The Institute of Internal Auditors and AIPCA & CIMA). In many cases, more than one of the following parties should be involved.

- While *software developers* are not sufficiently independent to perform audits for ongoing oversight, they should almost certainly be asked to provide initial testing outcomes along standardized measures before a model is sold or put into operation.

- NIST and traditional financial services recognize that an *internal but independent team* within the organization that operates a system should play a critical part in any sufficient oversight of that system via routinely scheduled and documented audits.

- An *external organization with a stake in the system's successful operation* (e.g., software-as-a-service ("SAAS") products that include support packages for oversight and

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system review) could play an important part in any comprehensive oversight.

- An *independent third party*–with "independent" varying in definition but essentially external and unconnected to the audited entity–is likely to play a central part in any regulatory oversight process. The NYC Law 144 appears to take this approach by clarifying that the auditor cannot have shared financial interests with the developer or operator of the automated system.

- *Regulators* are beginning to insert themselves into the oversight of automated decision-making systems. In all likelihood, regulators will only serve as actual auditors in cases where explicitly identified "high risk" systems are deployed at a vast scale and scope or when, per some policy, government auditors are brought in to review systems that failed their initial audit testing. Legislation might also require that government agencies review the audits conducted by some of the prior groups to confirm or spot-check compliance. Public sector resources are likely to a limiting factor in this area for the foreseeable future (Pablo Jiménez Arandia 2023).[14]

Second, *what should an audit include?* We have already said it is essentially an examination of the model's performance metrics, but this still leaves much room for discussion.[15] Traditional financial audits are scoped explicitly for clearly defined risks based on standardized definitions. This precision does not yet extend to AI audits. Even just limiting an audit to the examination of model outputs for "bias" is, ultimately, a value-laden choice. Determining which of the commonly used statistical tests (*e.g.* adverse impact ratio, t-test, percentage point difference, chi-square, Fisher's exact test, or others) are most appropriate for the context, how many should be run, and interpreting the results into understandable analysis are all decisions either the regulatory guidance must specify of the individual auditor must choose.

Furthermore, it is not a settled question whether or not to include a review and evaluation of system *data*, along with the system *outputs* discussed so far, and answers may vary based on circumstances. In some cases, it may be enough to review a model's performance solely by evaluating its outputs. However, in other cases, it may also be necessary–and perhaps legally mandated–to review the data. This task, however, is far from straightforward because there are different types of "data": training data is used to develop a model; testing data is used for model validation prior to deployment; and once a model is integrated into practice, it creates historical or real-world data. While it is comparatively easier to evaluate training and testing data due to

[14] Spain is the first EU country to create an oversight agency, and wants to lead the way in the regulation of Artificial Intelligence in Europe. Plans there include the creation of a "national AI seal," a certificate that accredits that the systems deployed in the country meet the requirements demanded by European law. This seal, however, will be voluntary, at least until the European regulation comes into force. The government is not clear about how it will encourage companies to undergo this recognition, but they do hint at the possibility of linking access to European funding for companies to this seal of quality.

[15] As noted above, it cannot be overemphasized that AI harms extend beyond bias in model design, data selection, and outputs. This audit discussion is to address the frequently highlighted concerns around embedded or learned bias as one of the high-profile risks of systems automated using historical data. But additional technical, social, and contextual harms must also be addressed in the general governance protections proposed or considered for these systems.

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their restricted operations in laboratory-like conditions, these constraints might not reveal real-world harms that arise when AI tools are deployed in the wild. Finally, there is the issue of auditing data sources. This requirement can be as controversial as making judgments about dataset accuracy, size, or diversity.

To illustrate the practical stakes of the "what to include" question, consider legislation that scopes out a broader mandate than NYC Local Law 144. A Washington, D.C. bill prohibits discrimination based on automated decisions impacting individuals' "major life opportunities," including, but not limited to, employment. (D.C. Bill 24-558 (2021). The bill requires service providers and any entity in possession of large amounts of personal information on D.C. residents to publish a notice to the person about the use of that information before making an AI-based determination about that individual. It also includes a requirement for mitigation and documentation to bring these systems into compliance.

The third question is, *what standard(s) should the audit outputs be measured against?* It is important to note that all systems now performing a function using AI are still subject to all the existing laws and regulations that already apply to their industry or business, whether sector-specific (such as financial services regulations) or broadly applicable (such as employment anti-discrimination rules). When existing legal or industry standards are available, such as in the decades of established analysis for disparate impact in employment, they can be applied. However, in many cases, established or clear legal standards may not exist. Determining what standard or measure is appropriate to the context or specific applications will not always be obvious or amenable to consensus.

The fourth question is *what should the output of the audit be, and who should have access to it?* An audit report can be structured in different formats, and it can include varying levels of detail about the model or data, the tests that were available or performed, the results and follow-up actions, and the results of any re-testing. The level of specificity may be influenced by considerations related to the target audience (e.g., regulators, the public, customers, or business partners), and it may even be necessary to craft different versions of reports required for those different audiences based on the audit's purpose and appropriate levels of confidentiality. It is important to note that detailed descriptions of auditing processes may reveal additional proprietary information about companies, business practices, competitive advantages, and vendor relationship status. It could even pose risks to consumers by revealing personal information. Thus, a decision that audit reports should be posted to the public will necessitate different levels of information and analysis than one provided in a protected status to a regulator, even if both are essentially intended to provide credible reassurance.

The fifth question is *how frequently should audits occur?* Most US laws and proposed bills anticipate an annual requirement for models in operational use. However, given the frequency with which software updates are made, and data sets are replaced or revised, a more appropriate standard might be to tie the periodicity to the risk level of the model. That is, models that carry higher risk should be audited more often, possibly twice a year or quarterly, as well as any time those significant aspects of the model change.

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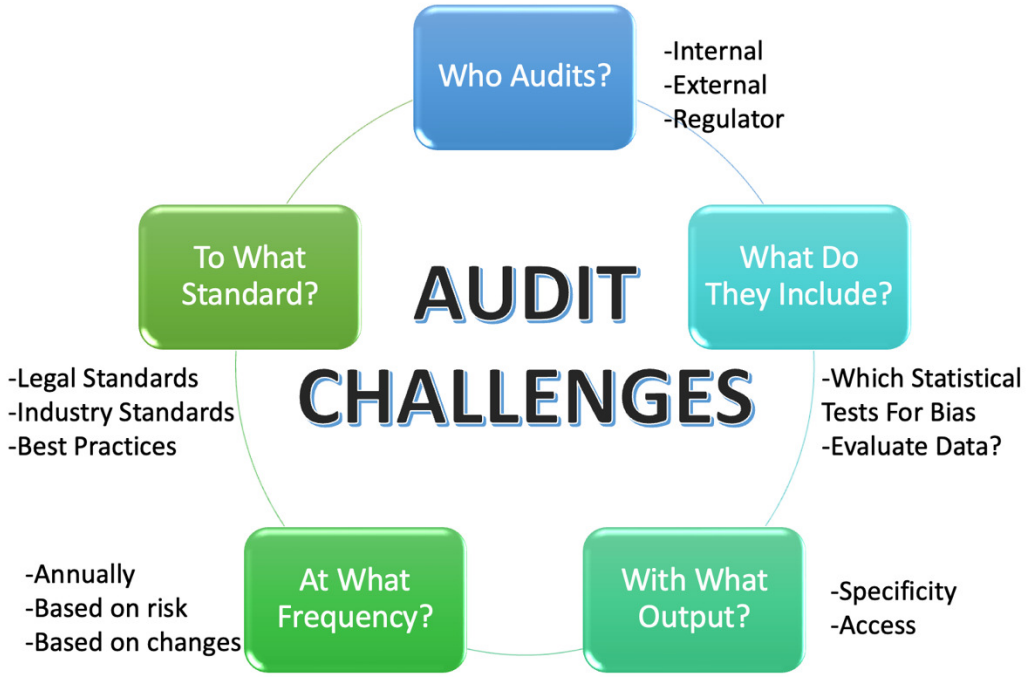

Applying These AI Audit Questions

To demonstrate the practical value of our main AI audit questions, we will apply them to the requirements of NYC Local Law 144. After revisiting that legislation, we will advance the analysis of conducting a reasonable AI audit by explaining how lawmakers can improve upon New York City's auditing process. So, as an example, the NYC law and proposed guidance answers to these five questions and an alternative approach based on our discussion and evaluation, might contrast as follows:

Question	NYC Local Law 144	Proposed Legislative Approach
Who:	An external independent auditor	Documented independent internal audits, confirmed by a professional/accredited external independent party.
Measuring What:	Two protected classes (gender and race), assessed independently and intersectionally	The appropriate bias measures based on context (*e.g.*, for employment, all protected classes).

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To What Standard:	Traditional disparate impact under established EEOC rules[16]	A specified, established legal standard where one exists; or else an industry standard, best practice, or defensibly "reasonable" measure based on industry and context
Results:	A publicly posted report of the audit outputs	• Developers/internal auditors – detailed findings documented within the company • External Auditors – summary level results, filed to the regulator • Consumers – public-facing posting of bottom-line outcomes where appropriate
Frequency:	Within a year prior to use (and therefore de facto, annually)	• High-risk systems – quarterly • Medium/Low Risk Models – semi-annually or annually • For major code revisions or significant updates or resets of data sources – upon event/occasion

Figure 3: Comparative Legislative Approaches to the Five Audit Questions

Because our case study involves a piece of existing legislation mandating an audit, we have framed these alternate answers as those appropriate for other legislators to consider in order to avoid some of the problems NYC has encountered. The NYC law was passed quickly—some might say too quickly—without having taken the time to work through some of the issues more carefully with expert and stakeholder inputs. Their requirements are fairly minimalist and too incomplete to achieve comprehensive oversight. Indeed, they may fall short of accomplishing the primary goal–which is to prevent additional bias in hiring and employment practices generally.

Therefore, we offer an alternative proposal. The next legislators might choose to require that companies do their own internal auditing along with specifying that external audits be performed either on a specified timescale or when triggered by other reports or factors that indicate a possibility of unfair impacts. While much of our discussion has focused on external audits based on legislative mandates, the reality is that—just as with financial audits—internal review and accountability play an integral role in responsible operations. Also, externally imposed requirements alone can never compensate for the ongoing oversight of a system's owners and operators.

When considering what to measure, these future legislators might focus on particular aspects of equity or fairness in carving out the specific groups or features to consider. And they

[16] While the NYC law is an evaluative requirement without a specified compliance standard, the implication is that the outputs will be measured against traditional impact ratio expectations.

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could identify existing legal standards or tie requirements to "industry practice" that can be adjusted over time. Thoughtful legislators will realize that while detailed findings should be available as and where necessary, that too much detail can compromise business interests and actually create vulnerabilities around the systems involved. Consequently, regulators should have access to sufficiently detailed but confidential reports to help ensure models are not causing harm. But they should frame public-facing results at a less granular level and with the aim of demonstrating greater transparency to generate trust. Finally, savvy legislators will recognize that a one-size-fits-all frequency of "annual" reviews is unlikely to be enough to promote truly responsive and responsible governance. They likely will need to be more nuanced when determining what audits they mandate and what time periods are appropriate.

However, it is also important to remember that our overall discussion is about the general value of audits as a governance tool, not merely as a regulatory mechanism. Therefore, who asks the guiding questions will influence what answers are given. The "askers" might include the following: a company seeking to responsibly develop AI systems for sale to other businesses; the businesses who purchase AI-driven products, services, features, or platforms for their own internal processes or as part of their customer-facing applications where consumers prioritize accountability; or a regulator charged with consumer protection or product safety responsibilities, including fairness, privacy, security or other digital challenges. In each case, the answers will vary based on the AI system involved, its risk level, and the intent and purpose of the audit. Thus, there is no one "right" set of answers for all times. Instead, the more challenging task is to carve out contextually appropriate AI audit requirements.

STOP the AI Audits

During the consideration of Local Law 144, civil rights and racial justice advocates raised questions that go to the heart of AI auditing's power and viability (Stop Surveillance Technology Oversight Project 2020). These objections are certainly notable for AI practitioners on the merits, raising concerns about nearly every aspect of the nascent technology. But even for those who disagree with these advocates' calls for AI moratoriums, holistic AI reviews, and aggressive private sector litigation, they still must confront what these objections say about the long-term viability of AI auditing. Even if lawmakers have historically been slow to respond to advocates' demands, AI firms must recognize the ongoing risk that a critical mass of public opposition could fuel more aggressive future laws, regulations, and enforcement actions.

First, advocates question if AI audits can truly be independent. (Boyle 2020, 56). This comes in part from a concern that even AI auditors who work for independent firms will face the same risk of institutional capture faced by accounting firms historically. Unlike the crippling financial penalties imposed by Sarbanes Oxley (and even the risk of imprisonment), Local Law 144, DC's proposed AI auditing law, and other local measures frequently impose minimal sanctions on auditors whose independence is compromised. Even when auditors work for external firms, those firms will be more likely to be hired by companies when they minimize the cost of compliance and rubber stamp a clean algorithmic bill of health. (West 2020, 67).

Additionally, while advocates agree with AI auditors on many of the factors that drive low-quality AI audits (such as lack of access to relevant data and lack of independence), they do

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not believe that solving these factors is enough to create a good audit (Stop Surveillance Technology Oversight Project 2021). That is because many of the harms that flow from AI systems are not found within the confines of the model itself, but the integrity of the data that model is used to analyze, the decisions made by human operators of an AI system, and other downstream drivers of bias.

Facial recognition provides a powerful example of the limited corrective power of AI audits. Facial recognition algorithms are frequently discriminatory against Black, Latinx, Asian, Female, non-binary, elderly, and very young individuals. (Damiani 2019; Buolamwini and Gebru 2018, 77-91). As a result, when a facial recognition scan is run against a person from one or more of these demographic groups, there is often a heightened risk of an incorrect result. But that algorithmic error is just one driver of discrimination. For police facial recognition, there is also frequently human bias in (1) which individuals' images are analyzed through facial recognition, (2) which faces that image is compared against, and (3) how the result of that comparison is treated. (Garvie 2019). As a result, even a successfully audited policing algorithm can drive deeply discriminatory outcomes. Civil rights groups warn that audits then could only serve to normalize and legitimize the discriminatory technology, addressing one, small subset of the system's overall biases. (Schwarz 2020, 7).

In addition to systems that advocates worry are audit-proof, like facial recognition, there are others that they fear are vulnerable to audit washing. (West 2020, 67). While there are many examples of companies that are successfully able to deploy audits in good faith to root-out bias, there are few (if any) stories of successful adversarial audits. (Goodman and Trehu 2022). Adversarial audits (conducted by an external firm, despite a client's opposition) can prove exponentially more difficult for auditors. This is both because of the potential conflict of interest to provide clients with favorable reports and because an adversarial client can provide far less access to relevant datasets. Advocates fear that audits conducted under these conditions not only could fail to identify discriminatory models, but that they could become both a legal and public relations shield from scrutiny. In this way, a low-quality audit could be more harmful than the absence of any audit at all.

Politically, audit washing can create several barriers to improved civil rights protections. First, ordinances like Local Law 144 can give political leaders and lobbyists a way to argue that the problem of discriminatory AI has been "fixed." Even before the measure took effect, lawmakers claimed that Local Law 144 would "protect NYC residents from bias and promote equal opportunities regardless of their gender or skin color." (New York City Council 2021). Such claims may age poorly as regulators struggle to finalize the law's terms. But it still remains a potent rhetorical response to advocates who call for government testing of algorithms, sectoral restrictions, and outright bans.

Additionally, audit washing can potentially be used as a defense in court. Not only are regulators struggling to understand how to implement the specifics of new AI laws like Local Law 144, but judges are faced with the task of deciding how to apply thousands of existing laws targeting human decision making to AI systems. Plaintiffs have already gone to federal court, claiming that discriminatory housing algorithms violate the Fair Housing Act, and other cases are likely to follow. (Kirchner 2020). And it's not just private litigants that see AI as potentially

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violating legacy civil rights laws. Last year, the Department of Justice and Equal Employment Opportunity Commission (2022) warned that AI tools can violate the Americans with Disabilities Act, creating barriers for job applicants and employees with disabilities.

But advocates fear that auditing mandates will be held up by companies as a way to evade liability under these existing laws. Companies can use the latitude afforded to them by lax auditing standards and regulations to certify that their software is nondiscriminatory. That may already be unfolding with voluntary audits, but auditing requirements may increase the risk. Companies can argue that auditing mandates suggest lawmakers intend for auditing to be the primary measure of AI discrimination. While Local Law 144 omits any language that would explicitly preempt existing civil rights laws, courts frequently interpret one statute as implicitly overturning a prior one. In this way, AI audits could transform from a floor on legal scrutiny to a ceiling.

Taken collectively, these concerns were enough to mobilize civil rights advocates against laws like Local Law 144. Instead of supporting audits as a regulatory mechanism, they've looked to more sweeping measures, such as banning the riskiest forms of AI.

Conclusion

Having covered many aspects of AI audits in this chapter, it will be useful to conclude by reviewing our main points. We began by providing a brief history of financial audits to demonstrate that audit standards and practices can evolve with a commitment to professionalization. Although financial audits are not perfect, they have become an essential tool for promoting financial responsibility. Crucially, while financial and AI audits are not the same processes, the former provides a good baseline for anticipating that the latter will improve.

Our discussion of Local Law 144 in New York City clarified why it is currently difficult to conduct even an apparently straightforward AI audit. Given the novelty of the process and a range of value-laden choices to be made, many issues concerning how to responsibly govern AI remain nuanced and challenging. Ultimately, while the law is a step in the right direction, it was passed far too quickly. Lawmakers should have been more patient and thorough, allowing more time to deliberate about definitions and determine rules. In fairness, however, these lawmakers are acting without clear standards or consensus on many issues. Consequently, it is not entirely surprising that the limits of this law feel more comparable to financial audits that took place before there were generally accepted accounting principles. Just as financial practices matured and financial audits became more sophisticated in response, AI standards will converge and solidify, and AI audits will consequently improve.

After this case study, we addressed the question of what constitutes a reasonable AI audit. To answer that, we began with three foundational claims that became the basis for five key questions. These questions serve as the template for appropriately designing audits for machine learning models that analyze patterns and perform a predictive function—not just the ones used to make employment decisions, like those targeted in the New York City law. We then applied those questions retrospectively to the New York City law and then offered contrasting answers to demonstrate how future legislation requiring AI audits could improve upon it.

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While most of the chapter presented AI audits as an achievable and crucial component of AI governance, we concluded by considering a skeptical position. Civil rights advocates not only argue that AI audits are insufficient to defend against the dangers posed by discriminatory systems, but fear that lax auditing mandates could insulate AI systems from future scrutiny by lawmakers and in court.

Our goal was to present an inclusive analysis of both the strengths *and* limitations of audits as a tool to manage AI systems because we believe it is crucial for practitioners to understand why they are subject to debate–to grasp why they are contentious, how they can be improved, and why some people remain deeply skeptical of their value. The process of auditing AI models will change over time, and preparing for the future of responsible AI governance requires anticipating invigorated attempts at creating best practices and common standards, as well as principled and passionate civil rights-based rejections of these endeavors.

Key Learnings

- People disagree on what constitutes an "audit" in the context of an AI model. We proposed a definition that aligns with our three foundational claims: it is a technical process that serves the public interest and determines whether the performance metrics comply with an existing standard as part of a larger AI framework. We do not expect everyone will agree with this definition because there are valid reasons for preferring alternatives. However, two especially valuable practical consequences follow from adopting it. First, since many practitioners use the definition, the shared vocabulary should increase the ease of communicating and coordinating behavior. Second, expecting too much from the auditing process can lead to misunderstandings about what is responsible for achieving and how those involved in the auditing process should meet their responsibilities. Minimizing misconceptions about purpose and purview might decrease some of the tensions that arise in the future as the AI audit process evolves and becomes more streamlined.

- Companies that need to comply with AI audit laws must develop their internal procedures for legal and machine learning teams to collaborate effectively, knowing that the legal rules will most likely be somewhat-to-extremely vague and allow conflicting interpretations. Proactive and flexible interpretations of the legal and technical requirements will require contributions from individuals with expertise in procurement, compliance, policy, and traditional risk assessment, as well as legal and technical skills. Larger and smaller organizations may require different approaches to create cross-functional teams or committees to prepare for and maintain compliance. Regardless of size, all organizations must have a clear accountability structure—policies and procedures that explicitly assign distinct responsibilities, including the necessary independence to challenge, pause, or halt specific use cases or sign off on audit results.

- AI developers and users should retain the flexibility to adapt to emerging regulatory requirements. Firms should recognize the geopolitical risk in deploying AI systems that may face stringent regulations or outright bans in some jurisdictions. While many

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proposals to date have been deferential to industry practices and norms, growing numbers of advocates continue to push for more sweeping and stringent intervention. As a result, firms that integrate AI broadly into key business processes may face high switching costs if faced with a more challenging regulatory environment. Additionally, companies must actively monitor how government regulators and private plaintiffs use existing civil rights laws to challenge AI systems.

- What role the government should play in AI auditing is an important question that requires more in-depth discussion. For example, it has been proposed that the Equal Employment Opportunity Commission should review and license some AI systems, such as algorithmic hiring tools, for pre-approval before companies use them, much like the Food and Drug Administration does with drugs. (Schellmann 2021). That level of involvement, however, is impractical. Not only would it be prohibitively expensive to create the necessary workforce, but the concept is inherently unworkable. Since AI models are constantly evolving, any approval could only be provisional and have a finite lifespan. Still, the EEOC could do a lot more than it is currently doing. One possibility, that Ifeoma Ajunwa (2023) raises, is to certify third-party vendors that offer audits to help ensure a level of quality performance. Another, that she asks us to consider, is to provide "its own automated governance tools in the form of AI or automated systems that could be used to audit automated hiring systems." (Ajunwa 2023).

Electronic copy available at: https://ssrn.com/abstract=4568208

References:

AICPA & CIMA 2023, Audit, Attest & Quality Control Standards. https://www.aicpa-cima.com/resources/landing/audit-attest-and-quality-control-standards

Ajunwa, I. 2023. Testimony of Dr. Ifeoma Ajunwa. *Meeting of January 31, 2023 - Navigating Employment Discrimination in AI and Automated Systems: A New Civil Rights Frontier*. U.S. Equal Employment Opportunity Commission. January 31. https://www.eeoc.gov/meetings/meeting-january-31-2023-navigating-employment-discrimination-ai-and-automated-systems-new/ajunwa

Arandia, P.F. 2023 What to expect from Europe's first AI oversight agency. *Algorithm Watch. https://algorithmwatch.org/en/what-to-expect-from-europes-first-ai-oversight-agency/*

Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., Pollak, S. D. "Emotional Expressions Reconsidered: Challenges to Inferring Emotion From Human Facial Movements," *Psychological Science in the Public Interest,* 20(1), 1-68. https://journals.sagepub.com/doi/10.1177/1529100619832930

Benjamin, R. 2019. *Race After Technology: Abolitionist Tools for the New Jim Code*. Cambridge, UK: *Polity Press*.

Bhaumik, D. and Dey, D. 2022. An Audit Framework for Technical Assessment of Binary Classifiers.

Brown, Lydia X. Z.; Shetty, Ridhi; Richardson, Michelle. 2020. Report – Algorithm-driven Hiring Tools: Innovative Recruitment or Expedited Disability Discrimination? *Center for Democracy & Technology*, December 3. https://cdt.org/insights/report-algorithm-driven-hiring-tools-innovative-recruitment-or-expedited-disability-discrimination/

Boersma, M., Manoorkar, K., Palmigiano, A., Panettiere, M., Tzimoulis, A., and Wijnberg, N. 2022. Flexible categorization for auditing using formal concept analysis and Dempster-Shafer theory. Cornell University, arXiv:2210.17330v1[cs.AI].

Boyle, C. 2020. Testimony of Christopher Boyle, Director of Data Research and Policy, New York County Defender Services. *Ethical Implications of Using Artificial Intelligence and Automated Decision Systems*. Committee on Technology. November 13: 56. https://legistar.council.nyc.gov/View.ashx?M=F&ID=8946634&GUID=4E3EC70F-E6FD-4F91-B863-ADFCD2A4ECE3

Buolamwini, J., Gebru, T. 2018. Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification. Proceedings of the 1st Conference on Fairness, Accountability and Transparency. *Proceedings of Machine Learning Research* 81:77-91, 2018. https://proceedings.mlr.press/v81/buolamwini18a.html.

Coeckelbergh, M. 2022. *The Political Philosophy of AI*. Boston, MA*: Polity Press*: 42-48. Cornell University, arXiv:2211.09500v1[cs.CY]. .

Damiani, J. 2019. New Research Reveals Facial Recognition Software Misclassifies Transgender, Non-Binary People. Consumer Tech. *Forbes*. October 29. https://www.forbes.com/sites/jessedamiani/2019/10/29/new-research-reveals-facial-recognition-software-misclassifies-transgender-non-binary-people/?sh=7375c592606b.

European Commission. 2022. The Digital Services Act. https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/europe-fit-digital-age/digital-services-act-ensuring-safe-and-accountable-online-environment_en#documents

Employment Opportunity Commission. 1979. Questions and Answers to Clarify and Provide a Common Interpretation of the Uniform Guidelines on Employee Selection Procedures. *Federal Register* 44, no. 43. March 2**:** 11729-12015.

Garvie, C. 2019. Garbage In, Garbage Out. Center for Privacy & Technology. *Georgetown Law*. May 16. https://www.flawedfacedata.com/.

Goodman, E. P. and Trehu, J. 2022. AI Auditing and Accountability. *The German Marshall Fund* November: 3. https://www.gmfus.org/sites/default/files/2022-11/Goodman%20%26%20Trehu%20-%20Algorithmic%20Auditing%20-%20paper.pdf.

Hasan, A., Brown, S., Davidovic, J., Lange, B., and Regan, M. 2022. Algorithmic Bias and Risk Assessment: Lessons from Practice. *Digital Society* 1 (1): 4.

International Organization for Standardization (ISO). 2016. *ISO 9001 Auditing Practices Group Guidance on Impartiality*, International Organization for Standardization, Auditing Practices Group: GPO. https://committee.iso.org/files/live/sites/tc176/files/documents/ISO%209001%20Auditing%20Practices%20Group%20docs/General/APG-Impartiality2015.pdf (Accessed March 10, 2023).

Electronic copy available at: https://ssrn.com/abstract=4568208

Jones, J., Frazier, J., and Koerner, K. 2023. *Privacy and AI Governance*. International Association of Privacy Professionals. https://iapp.org/media/pdf/resource_center/privacy_ai_governance_report.pdf

Joyce, E. J. 1976. Expert Judgment in Audit Program Planning. *Journal of Accounting Research*, vol. 14: 29–60. https://doi.org/10.2307/2490445.

Kirchner, L. 2020. Can Algorithms Violate Fair Housing Laws? Locked Out. *The Markup*. September 24. https://themarkup.org/locked-out/2020/09/24/fair-housing-laws-algorithms-tenant-screenings

Maurer, R. 2021. New York City to Require Bias Audits of AI-Type HR Technology, *Society for Human Resources Management*, December 20. https://www.shrm.org/resourcesandtools/hr-topics/technology/pages/new-york-city-require-bias-audits-ai-hr-technology.aspx.

National Institute for Standards and Technology. 2023. *Artificial Intelligence Risk Management Framework*. Washington D.C.: U.S. Department of Commerce, January 26. https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.100-1.pdf.

Narayanan, A. 2018. Tutorial: 21 Different Definitions of Fairness and Their Politics. YouTube. https://www.youtube.com/watch?v=jIXIuYdnyyk.

New York City Department of Consumer and Worker Protection. 2023. *Notice of Adoption of Final Rules https://rules.cityofnewyork.us/wp-content/uploads/2023/04/DCWP-NOA-for-Use-of-Automated-Employment-Decisionmaking-Tools-2.pdf*. New York City: NYC Rules. (Accessed April 12, 2023).

New York City Council. 2021A. *Council Votes to Overhaul Construction Safety*. November 10. https://council.nyc.gov/press/2021/11/10/2111/.

New York City Council 2021B. *A Local Law to Amend the Administrative Code of the City of New York, In Relation to Automated Employment Decision Tools.* https://legistar.council.nyc.gov/LegislationDetail.aspx?ID=4344524&GUID=B051915D-A9AC-451E-81F8-6596032FA3F9&Options=ID%7CText%7C&Search=.

Office of Justice Programs, U.S. Department of Justice. 2022. *Generally Accepted Accounting Principles (GAAP): Guide Sheet*, The Office of Justice Programs Territories Financial Support Center (OJP TFSC), Office of Justice Programs, Financial Management Resources and Tools, Washington, D.C: GPO. https://www.ojp.gov/sites/g/files/xyckuh241/files/media/document/GAAP_Guide_Sheet_508.pdf (accessed March 10, 2023).

Levy, H. B. 2020. History of the Auditing World, Part 1. *The CPA Journal.* November. https://www.cpajournal.com/2020/11/25/history-of-the-auditing-world-part-1/.

Lobel, Orly. 2022. *The Equality Machine: Harnessing Digital Technology for a Brighter, More Inclusive Future*. New York, NY: *Public Affairs*: 49-52.

Schaake, M. and Clark, J. 2022. Stanford Launches AI Audit Challenge. Stanford University, Human-Centered Artificial Intelligence, *Law, Regulation, and Policy* July, 11. https://hai.stanford.edu/news/stanford-launches-ai-audit-challenge.

Schellmann, Hilke. 2021. Auditors Are Testing Hiring Algorithms for Bias, But There's No Easy Fix. *MIT Technology Review*, February 11. https://www.technologyreview.com/2021/02/11/1017955/auditors-testing-ai-hiring-algorithms-bias-big-questions-remain.

Schmidt, J. 2023. "Legal Liability of Auditors," *CFI*. January 5. https://corporatefinanceinstitute.com/resources/accounting/legal-liability-of-auditors/.

Schwartz, D. 2020. Testimony of Daniel Schwart On Behalf of the New York Civil Liberties Union. *Ethical Implications of Using Artificial Intelligence and Automated Decision Systems*. Committee on Technology. November 13: 7. https://legistar.council.nyc.gov/View.ashx?M=F&ID=8946634&GUID=4E3EC70F-E6FD-4F91-B863-ADFCD2A4ECE3

Society for Human Resource Management. 2022. *Automation & AI in HR*, Society for Human Resource Management, February. https://advocacy.shrm.org/SHRM-2022-Automation-AI-Research.pdf?_ga=2.112869508.1029738808.1666019592-61357574.1655121608.

Solove, D. 2023. Data Is What Data Does: Regulating Use, Harm, and Risk Instead of Sensitive Data. *Northwestern University Law Review* 118, January 11. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4322198.

Stop Surveillance Technology Oversight Project. 2020. RE: Int. 1894-2020 - Sale of Automated Employment Decision Tools. https://www.stopspying.org/latest-news/2020/11/13/re-int-1894-2020-sale-of-automated-employment-decision-tools

Stop Surveillance Technology Oversight Project. 2021. Letter to Council Member Cumbo on Amending Intro 1894-2020. https://www.stopspying.org/sign-on-letters/2021/4/22/letter-to-council-member-cumbo-on-intro-1984-2020-the-need-for-a-moratorium

The Institute of Internal Auditors. About Internal Audit. https://www.theiia.org/en/about-us/about-internal-audit/

Electronic copy available at: https://ssrn.com/abstract=4568208

U.S. Equal Employment Opportunity Commission. 2022. *2021 EEO-1 Component 1 Data Collection Instruction Booklet*. U.S. Equal Employment Opportunity Commission. Washington, D.C.: GPO. https://www.eeocdata.org/pdfs/2021_EEO_1_Component_1_InstructionBooklet.pdf.

--. 2022. *The Americans with Disabilities Act and the Use of Software, Algorithms, and Artificial Intelligence to Assess Job Applicants and Employees*. U.S. Equal Employment Opportunity Commission. Washington, D.C.: GPO. https://www.eeoc.gov/laws/guidance/americans-disabilities-act-and-use-software-algorithms-and-artificial-intelligence

U.S. Securities and Exchange Commission. 1998. *Policy Statement: The Establishment and Improvement of Standards Related to Auditor Independence*, by Jonathan G. Katz, No. 33-7507; 34-39676; IC-23029; FR-50, Washington, D.C. https://www.sec.gov/rules/policy/33-7507.htm.

Verma, P.. 2023. AI is starting to pick who gets laid off. *The Washington Post*, February 20. https://www.washingtonpost.com/technology/2023/02/20/layoff-algorithms/

Webber, M. R. 2022. The Enron Scandal and the Sarbanes-Oxley Act. *the balance*. September 13. https://www.thebalancemoney.com/sarbanes-oxley-act-and-the-enron-scandal-393497.

West, S. 2020. Written Testimony of Dr. Sarah Myers West, Postdoctoral Researcher, AI Now Institute. *Ethical Implications of Using Artificial Intelligence and Automated Decision Systems*. Committee on Technology. November 13: 67. https://legistar.council.nyc.gov/View.ashx?M=F&ID=8946634&GUID=4E3EC70F-E6FD-4F91-B863-ADFCD2A4ECE3

Wilson, C. 2020. Auditing Algorithms. Northeastern University, Khoury College of Computer and Information Sciences. https://personalization.ccs.neu.edu/.

Zhang, B. H., Lemoine, B., and Mitchell, M. 2018. Mitigating Unwanted Biases with Adversarial Learning. Cornell University, arXiv:1801.07593v1[cs.LG]. January 22. https://arxiv.org/abs/1801.07593.

Electronic copy available at: https://ssrn.com/abstract=4568208